Weighted Weil-restriction experiments on larger curves

2026-10-19

Observed results

The native primary suite and larger-recovery follow-up completed 36 of 36 synthetic one-target DLP recoveries on 12 distinct prime-order curves over cubic extension fields. Base primes range from 31 to 1009; the largest group order is 1,027,286,801. No recovery budget was exhausted. The solver constructs no covering curve: it enumerates one weighted signed slot and uses native S3 resultant/root solving for the residual pair on the Weil restriction. The coefficient arms 1, 2 and 4 share the same factor-base logarithm unknowns.

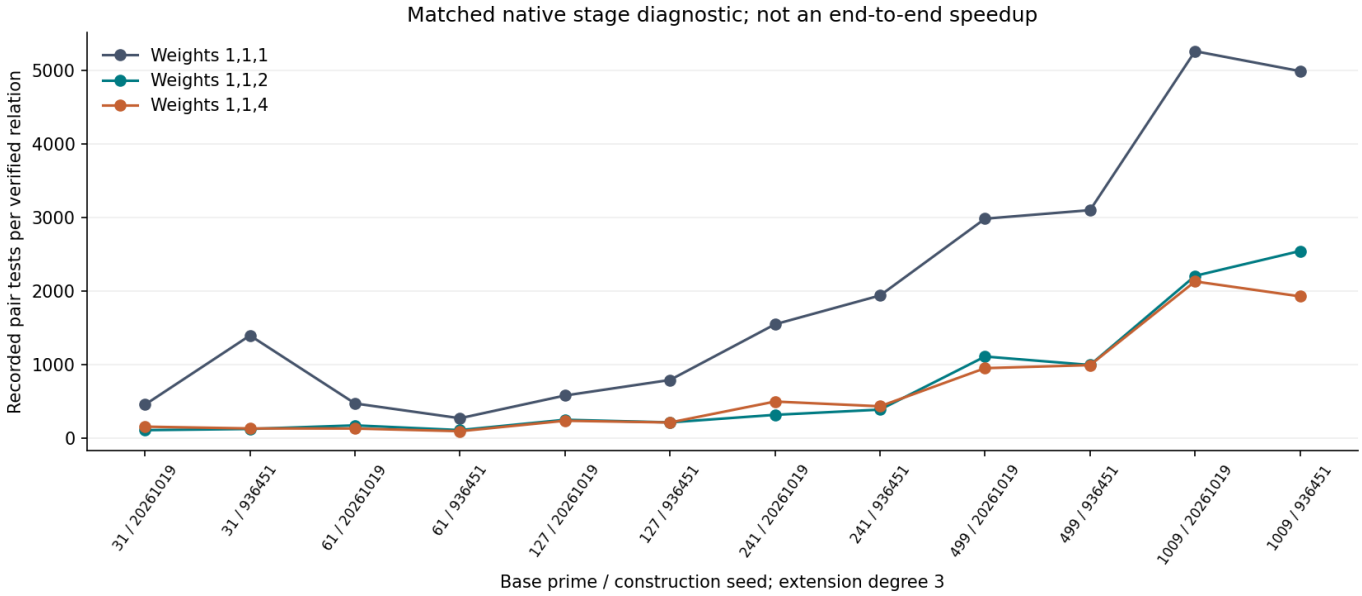


Figure 1: Primary run: native-20261019-primary; 128 matched targets per curve and arm. Values are recorded algebraic pair queries per verified decomposition. No uncertainty interval was computed.

On the two p=1009 curves the following stage yields and full-recovery residual counts were observed. The residual count does not price setup, polynomial arithmetic, group operations or linear algebra.

Seed	Weight c	Stage relations / 128	Full-recovery residuals
20261019	1	21	3044
20261019	2	45	1247
20261019	4	47	1152
936451	1	22	3063
936451	2	42	1302
936451	4	52	1177

Validation and evidence

The primary run recorded 11,370 PDP dictionary checks; the exploratory follow-up recorded 18,344. These are validation calls and include A/A and overlapping stage panels. Every returned decomposition was checked in the group. Every recovered scalar passed both a group check and the external known-answer check; the solver instance had its secret field set to zero. The independent signed-pair dictionary is unavailable to the algebraic solver. It is a quadratic correctness reference, not a proposed scalable relation-collection algorithm.

The primary replay reproduced every per-case result and deterministic counter byte for byte, including eight baseline DLP A/A controls. Its 24 duplicate recoveries are excluded from the 36 distinct cases; see control arrangement. Native PARI 2.17.2 separately confirmed all 12 curve cardinalities, generator orders and planted targets, and repeated those checks on the four follow-up fixtures. This is independent fixture arithmetic, not independent-host implementation replay.

The primary protocol was committed before measurement at 6095144c, with the primary runner at fc13c52e. The follow-up protocol and runtime source were frozen at 040b8e1a before execution. The follow-up was specified after seeing the primary results and uses the same four larger curves and stage panels. It is exploratory and adds 12 full recoveries. The primary/replay/follow-up reports have their exact commands, source snapshots, host information, budgets, raw-record links, graphs and PDFs:

- Primary run report.
- Deterministic replay report.
- Larger full-recovery report.
- Validation logs and initial failures.

Scope and next experiment

Observed weighted-slot yield gains survive this larger native suite. The number of unknown logarithms does not increase because the weight is a known scalar multiplying an existing base log. Breaking permutation collisions is a plausible explanation; this suite does not isolate that mechanism or establish publication novelty or an asymptotic gain.

Complete S, IC/rho ratio and end-to-end speedup remain null. The legacy pair API has incomplete native counters, its field-inverse accounting is legacy, and no complete conversion band or matched ecbench strong-rho reference was measured. Magma, msolve and FGb were not run. There is no independent-host replay. These are L0 count diagnostics on a shared AMD EPYC 9V74 KVM container with a four-CPU quota and 16 GiB memory; wall time is not admitted. Earlier Python results are historical and are not native runtime baselines. No cryptographic-size or binary-field claim follows.

The tracked gap specifies the next discriminating experiment: complete native instrumentation and a matched cold, one-target ecbench IC/strong-rho comparison on these frozen public fixtures, followed by independent-host replay.